

AGE ESTIMATION FROM FACE IMAGE LEVERAGING CONCATENATED FEATURES OF VISION TRANSFORMER ALONG WITH RESNET-50

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Abstract—Age is a biometric trait used in various applications such as recommendation, access control, criminal detection, so on. Age estimation algorithms primarily rely on facial features, with deep learning often showing promising results in this domain. However, due to the wide range of faces and numerous other extrinsic and intrinsic features, biological variances, and uncertainty make it complicated to estimate age. Thus, a single deep learning-based approach may downplay the significance of some extensive face data and other facial age features. Deep residual networks and Vision Transformer models are commonly used for facial age estimation, but their combined effectiveness and contrast in estimating human chronological age have not been thoroughly examined. To address these challenges, this work proposes a novel hybrid model combining the strengths of two state-of-the-art architectures, ResNet50 and Vision Transformer (ViT), capture local and global contextual information, improving age estimation robustness which outperforms standalone ResNet50 and Vision Transformer models, achieving state-of-the-art accuracy. The analysis evaluates the effectiveness of multifeature fusion from face images, including HOG-ResNet50-ViT features, comparing different machine learning and deep learning algorithms. The proposed method utilizes the UKTFace dataset, involving 23708 images, and attains 4.88 MAE by addressing age estimation as a regression problem.

Index Terms—Age Estimation, Facial Image, Vision Transformer (ViT), ResNet50, Features Fusion.

I. INTRODUCTION

Researchers' study facial features for age, gender and ethnicity identification. Age estimation from face images is challenging for computer vision applications because of changes in facial features brought on by medical conditions, cosmetic treatments, and other circumstances. Differences in appearance can be brought about by factors such as sunshine, smoking, gravity, and bone remodeling. Due to restrictions on publicly available age data, accurate face data is difficult to obtain. Advancements in computer vision and machine learning have improved our ability to infer age from physiological and behavioral markers. These techniques rely on manually created features, with some researchers combining handcrafted [2] features and Convolutional neural networks (CNNs), employ

hierarchical structure to encode both local and global data. However, the more complex architecture of CNNs results in constraints on locality and weight sharing and inductive biases. Deep learning (DL) models were enhanced by researchers using novel techniques to increase performance and practical applicability such as Vision Transformer [33], which builds linkages between visual parts without regard to established position or weight-sharing restrictions by using self-attention processes to weight the relevance of each component of the input image. This flexible approach captures both local and global relationships, enabling the model to, if necessary, concentrate on remote locations. We seek to discover and offer a high-performance model for face age estimate based on the thorough examination and comparison, which may be an invaluable resource in a variety of real-world applications. The main contribution of this work are as follows:

- This work suggests a novel age estimation technique by fusing the features from prominent ResNet-50 and Vision Transformer (ViT), deep learning models, which demonstrate comprehensive performance than the standalone models.
- We empirically demonstrate that our model performance is marginally better than the approaches consider the age estimation problem in a regression through classification manner.

The rest of the paper is organized as follows. Section II briefly reviews related work for age estimation methods. Section III describes the proposed ResNet50-ViT age estimation method. Section IV presents experimental results and analysis leading to conclusions in Section V.

II. RELATED WORK

Age estimation research impacts biometrics, surveillance, and human-computer interaction, potentially affecting the marketing, healthcare, and law enforcement sectors. Large-scale face image datasets and computer vision techniques influence to develop algorithms for age estimation. This section explores

previous research on feature extraction and selection techniques for face image processing, focusing on the complexities of age prediction from faces for real-world applications. Facial age identification is divided into two ways based on feature space and feature extraction methodology: appearance-based feature extraction (Global Features) and geometry-based feature extraction (Local Features). Initially, researcher works on some anthropometric models [5], active appearance models [6], texture-based models [7] using different handcrafted facial features extraction methods to estimate age from face. Feature extraction was carried out using the Gabor wavelet (GW) [1], local binary patterns (LBP), local phase quantization (LPQ), histograms of oriented gradients (HOG) [2] and integrated using a feature fusion approach, and estimated the ages and age groups using SVM and SVR [2]. M. M. Sawant et al. [3] developed a hierarchical multi-class Gaussian framework for automatically estimating age based on group-specific aging trends. D. Lu et al. [4] proposed a new age estimation methodology that integrates Gabor feature fusion with an improved atomic search algorithm, using a statistical histogram to encode and combine Gabor scale features, CIASO-SA for feature selection, and SVM for age categorization. I. H. Casado's [8] research on feature fusion techniques, including HOG, LBP, and SURF, demonstrated a 4.25-year improvement in MAE scores compared to handcrafted BIF.

CNN-based methods for age estimation have gained widespread adoption due to their superior performance compared to statistical regression and classification methods. Cao et al. [13] proposed the CORAL framework ensuring consistent rank ordering and reliable confidence scores using ordinal regression-based techniques, independent of architecture. Y. et al. [9] proposed another algorithm that enhances age categorization by considering age labels' relationships and minimizing gaps, using 20 Deep ConvNets and 10 patches for training and integration. M. Akhand et al. [27] proposed a pipeline technique for age estimation from human facial photos, utilizing transfer learning to control pre-trained deep CNNs. H Liu et al. [10] proposed ODFL and ODL, which learn features from raw pixels, making them more resilient to face expressions, size ratios, and complex backgrounds. A combination of Convolutional Neural Network (CNN) and Support Vector Regression (SVR) [11], pre-trained CNNs like VGG, Res-Net, Google-Net, Alex-Net to identify age gaps [17] are some example to employ CNN in age estimation. Fariza et al. [18] compare Resnet and ResNeXt to improve image categorization accuracy. S. H. Nam et al. [16] proposed a deep convolutional neural network-based age estimate method to reconstruct low-resolution face images, overcoming the limitations of low-resolution cameras and close-up face shots. To focus on gender influence on age estimation, a study by Hen et al. [14], and Gupta et al. [15] compare classical and deep learning methods and models. However, Rizwan et al. [12] utilized Recurrent Neural Network (RNN) in age and expression recognition but found it struggled to accurately identify individuals from distant photos based on facial features. The Attention mechanism [20] aggregates

input sequences using inner products, calculating aggregation weights. It enhances inference capability for sequence-to-sequence and sequence-to-one difficulties. The encoder maps input sequences onto higher-dimensional spaces, minimizing visual clutter through task-informative picture signals. Transformers, a novel formulation of attention-based stacked layers, allow encoding sequences without RNN layers [21]. Its encoders and decoders use Multi-Head Attention and Feed Forward layers to determine sequence components' relative location and order, a technique used in various computer vision tasks [22]. Our work has been inspired by attention-based single-sentence classification challenges in NLP. Some of the face-based age estimation approaches are Attention-based Dynamic Patch Fusion (ADPF) [19] and dual image augmentation aggregation (Transformer Encoder) [23]. Both models are designed for greater accuracy across ages. Our work, motivated by other works, investigates, analyses, and compares various approaches, models, and methods for accurately estimating an individual's age from their facial features, with a focus on the use of pre-trained CNNs and Vision Transformer (ViT). HOG features are also used to compare the relative efficacy of deep learning and machine learning-based methods for age estimation.

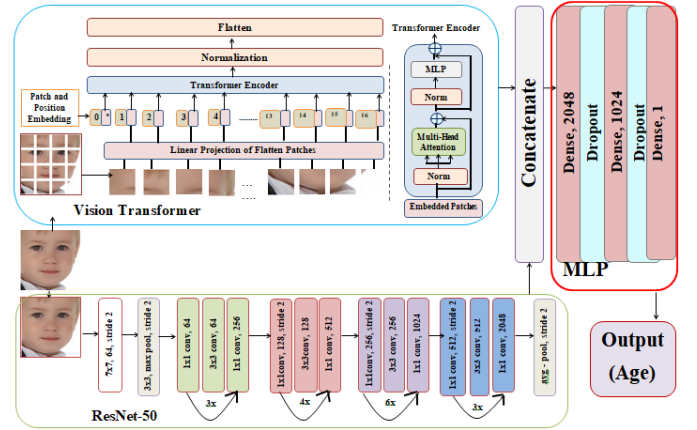


Fig. 1. Illustration of the proposed architecture

III. METHODOLOGY

The proposed hybrid model, vision transformer (ViT) and resnet50 feature concatenation model, architecture for age estimation is discussed in this section. ViT's exceptional performance in image processing tasks serves as a testament to its potential, and the CNN-based model is evaluated for the lowest error in age estimate. HOG is used to assess how well classic hand-crafted characteristics and contemporary abstract features work together. Fig. 1 shows the proposed model architecture whereas Fig 2 depicts the whole workflow of the methodology.

A. Facial Image Preprocessing

Human face detection is complicated work as there is a vast variety of appearance and external and internal factors. Here

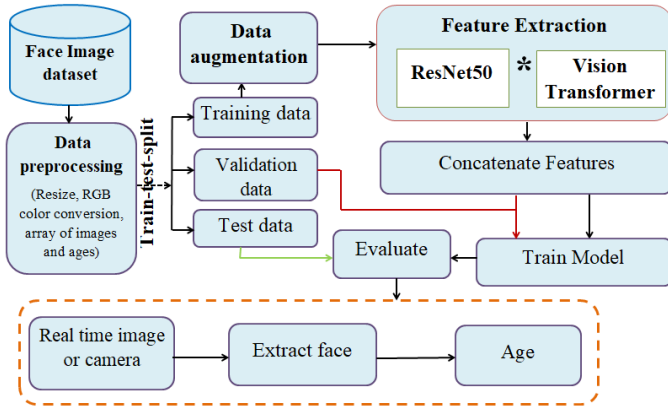


Fig. 2. Workflow diagram of the proposed method

are the utilized methods for preprocessing facial images of our work is given.

1) *Face Detection and Alignment*: Face detection is crucial for determining age and identifying interest regions. UTKFace's face alignment-cropped datasets from Kaggle are used for training, and validation time. Multi-task Cascaded Convolutional Networks (MTCNN) [34], a multi-task learning framework which works face detection and alignment tasks simultaneously. MTCNN.detect_faces() function detect and align the face efficiently and is used while testing with real-time images. The algorithm then selects faces' spatial coordinates for annotations after aligning them with landmarks. The work also separately preprocessed the dataset to meet each pre-trained CNN model's specifications, like rescaling, converting to BGR, and shrinking to 224x224 pixels.

2) *Data Augmentation*: Data augmentation is a technique that artificially expands a training dataset to alleviate overfitting, improve model performance, and reduce costs and labor-intensive procedures. In this work, an imbalanced dataset was used which bias the model's performance. So, from TensorFlow's Keras API, random flipping was set to horizontal flip, reducing the likelihood of the model recognizing upside-down images. Random rotation was adjusted to 0.2, and random enhancements were added to photos before submitting results to the training model. Fig 3 depicts some facial image after applying data augmentation on a image:



Fig. 3. Data Augmentation on UTKFace Dataset

3) *Regression and Classification*: Regression is a method used to establish the link between input and output variables in continuous data. Classification algorithms like random forest, decision trees, logistic regression, SVM are used for explicit classification. Age detection is a regression problem. However, age regression training can lead to uncertain predictions due

to instability in handling outliers [25], [35]. This study uses a hybrid model to estimate age from face images, utilizing the "softmax" algorithm to convert data into probabilities for anticipated age classes. The best-estimated age is obtained from both linear regression and classification, ensuring appropriate class instruction and balanced distribution.

B. Age Estimation Models

In our work, age estimation is done with utilizing different type of feature vectors such as from deep learning resnet50-ViT and HOG features from handcrafted feature vector. Each of this features vector are evaluated not only individually but also in combined mode. By evaluating each mechanism, our proposed resnet50-ViT gives highest performance result.

1) *Convolution Neural Network (CNN) Utilization*: Without human interaction [24], modern CNN-driven algorithms extract features, sometimes needing complicated layers of complexity. Backpropagation methods are used by CNN with convolution, pooling, fully connected layers to learn hierarchical spatial information. In our study, we used numerous pre-trained CNNs to estimate an individual's age only from a single facial image. These networks output an estimated age after receiving as input a face that was aligned with its background. Our approach made use of the widely used different pre-trained CNN models, picked for their ability to be managed in depth, their success in the ILSVRC in various years and their ability to balance performance and prediction speed. The best CNN model is ResNet50, which allows for the training of a huge number of layers without sacrificing speed [18]. ResNet50's design has a residual block for gradient movement and a skip link that allows input to traverse levels without layer weights.

2) *Vision Transformer Utilization*: A deep learning architecture called the Vision Transformer (ViT) locates spatial correlations in computer vision by using self-attentional processes and embeddings to record positional data. When compared to CNNs, it may be less able to capture fine-grained information because it relies on patching the image together. While ViT model makes use of attention to recognize context and long-range relationships, ResNet50 catches low-level features like edges, textures, and shapes. A powerful ensemble solution for age estimate from face images is provided by the combination of Vision Transformer and ResNet50. This method's effectiveness in capturing both local and global factors is what makes it work. Age estimate is precise and adaptable because to the hybrid model design, which consists further of two dense and dropout layers. Fig 1 shows our proposed hybrid model architecture.

3) *HOG Features Utilization*: HOG features are a type of computer vision technology used to find items in images. It also estimates facial age using histogram and gradient orientation computation on face spatial areas. The resulted HOG features are used as input for machine learning model like SGD Classifier, Random Forests and so on to estimate age [32]. In our work, we use HOG features extracted from the whole image of size 224*224 with the 'cv2.HOGDescriptor'

from 'opencv2', block-cell size of 8*8, and a block stride of 4*4, compressed into 9 vectors (each from 0 to 360 degrees with 40-degree bins). The main idea behind is that intensity gradients are used by the HOG descriptor to describe local object shape and appearance in pictures. For each cell's pixels, a histogram of gradient axes is generated, creating a descriptor.

4) *Concatenated Features Utilization*: The ViT-based features and the CNN-based abstract features (see Fig. 1) are concatenated into one new feature vector, then a terminal fully connected (FC) layer, a lineal layer for the final decision, was used. Moreover, while doing regression through classification, the terminal FC layer use 'softmax' activation function. The study first extracted the features from face image and applied different ML and DL algorithms. In case of DL, we used a final FC layer's containing 1 nodes (regression) and final FC layer's containing N nodes chosen based on the number of ages in the dataset (regression through classification). The regression method considered mean absolute error as performance metrics. The classification method took each age as an individual class label, encoded by using LabelEncoder.fit_transform() function and used 'categorical_crossentropy' as loss function. The same principle are applied to the feature concatenation of HOG-Resnet50's features, HOG-Resnet50-ViT features to find which features combination gives lowest error in age estimation.

C. Dataset

The study uses UTKFace [26], a large dataset of over 23708 face images (0-116) to train a facial age estimation. The dataset includes various poses, facial expressions, lighting, occlusion, and resolution details. Age, gender, and racial details are listed next to each image name. The dataset is divided into training, testing, and validation tasks, with training data comprising 15172 samples, testing data 4742 samples, and validation data 3794 samples. Fig 4 shows some sample images from UTKFace dataset and Fig.5 depicts the training data distribution:



Fig. 4. UTKFace dataset

IV. EXPERIMENTAL RESULTS

A. Experimental Setup and Parameter Settings

In this study, TensorFlow, Keras was used for network training, with Google Collaboratory GPU providing 12GB RAM. The network used an RGB facial picture as input and an Adam optimizer with a learning rate of 0.001, weight decay of 0.0001. The vision transformer had 12 heads, 28*28 patch

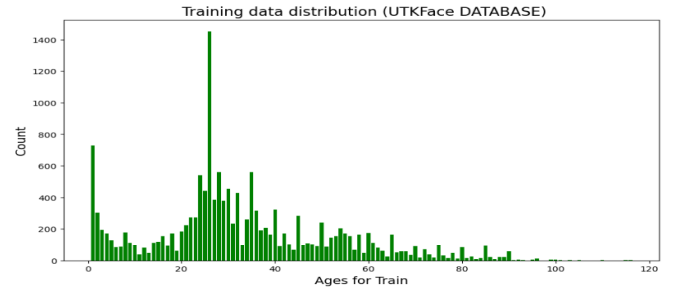


Fig. 5. Age distribution of UTKFace

sizes, 32 dimensioned feature spaces, 40 stacked layers, kernel initializer of MultiHeadAttention layer was 'glorot-uniform' with dropout of 0.1 and 'gelu' activation function with a batch size of 32. ModelCheckpoint and EarlyStopping were used to monitor validation loss over 7 epochs.

B. Performance Metric

Our technique for estimating age from face was evaluated using Mean Absolute Error (MAE), a standard metric, for both positive and negative evaluations.

1) *Mean Absolute Error (MAE)*: MAE computes the error between the estimated ages and the actual ages. The equation to calculate MAE is:

$$MAE = \frac{1}{N} \sum_{i=1}^N |A_i - P_i| \quad (1)$$

where N= number of test, A_i = Actual age, P_i = Estimate age.

2) *Accuracy*: Accuracy metrics is used to show the performance of model. The equation to find accuracy is:

$$\frac{T_p + T_n}{T_p + F_p + F_n + T_n} \quad (2)$$

Here T_p = True Positive, T_n = True Negative, F_p = False positive, F_n = False Negative.

C. RESULTS

This study aims to estimate age using facial features. The age estimation result that we have reported has the form of MAE. To determine a person's real/biological age, we tested our algorithm using the UTKFace dataset. Table I displays the average absolute error values obtained from a large-scale experiment using popular off-the-shelf deep learning models. The comparison table makes it very clear that combining the properties of well-known deep learning models, ViT and Resnet, results in considerable better performance than the standalone models. The results in TABLE I and Fig.6 shows that the proposed method provides with a mean absolute error (MAE) of 4.886 when using age as a regression method.

Table II presents a performance comparison between our approach and the current methods in terms of the mean absolute error achieved on the UTKFace dataset. Compared to the efforts of CORAL [13], ResNeXt-50 [18] and others' (See Table II), we achieved improved MAE.

TABLE I
EXPERIMENTAL RESULTS

Types of Features	Regression		Regression through Classification	
	Model	MAE	Model	MAE
HOG	DNN	7.085	DNN	8.125
CNN	ResNet50	5.021	RestNet50	18.058
ViT	—	7.870	—	9.705
HOG * ResNet50	DNN	6.082	DNN	6.104
HOG * ResNet50 * ViT	DNN	5.812	DNN	5.541
ViT * ResNet50	Proposed hybrid model	4.886	Proposed hybrid model	6.057

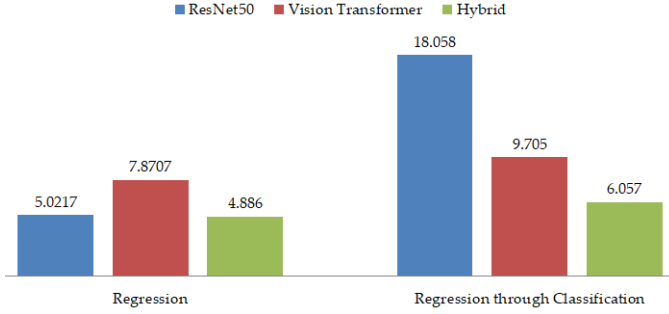


Fig. 6. Comparative Performance Of Proposed Hybrid Model

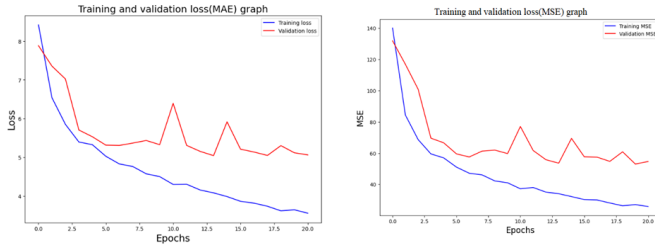


Fig. 7. Training And Validation Graph Of Proposed Hybrid Model



Fig. 8. Predicted Age Of Some Test Images Using The Proposed Model

V. CONCLUSION

In this paper, we have proposed a simple but effective hybrid model to estimate age from facial image. Our proposed model have designed to concatenate feature of two different deep learning model: Resnet50 and Vision Transformer. The work used machine learning and deep learning methods to estimate age and gender using the UTK-Face dataset, which consists of 23,708 photos. With a lower MAE of 4.88 on the UTKFace dataset, the proposed hybrid model shows promise for real-world age prediction by extracting facial images from real-

TABLE II
COMPARISON FOR AGE ESTIMATION ON UTKFACE DATASET

Method	Age Estimation(MAE)
Linear Regression [18]	11.73
ResNet50 [27]	9.66
ResNeXt-50 (32x4d) [18]	7.21
Inceptionv3 [27]	9.50
DenseNet [27]	9.19
Custom CNN [28]	5.67
AGES [29]	6.77
SVM [30]	7.25
CORAL [13]	5.39
EfficientNet-B3 [31]	6.71
Proposed hybrid model	4.88

time webcam or security camera recordings. The method can be modified to estimate age in practical situations. The study mainly sought to estimate age with minimal error (MAE) between actual and anticipated age. Layers, neurons, and learning rate can all be changed to create improvements. The study of others face image databases, racial and gender effects, and biometric indicators like voice, fingerprint, and handwriting will all be explored in the future.

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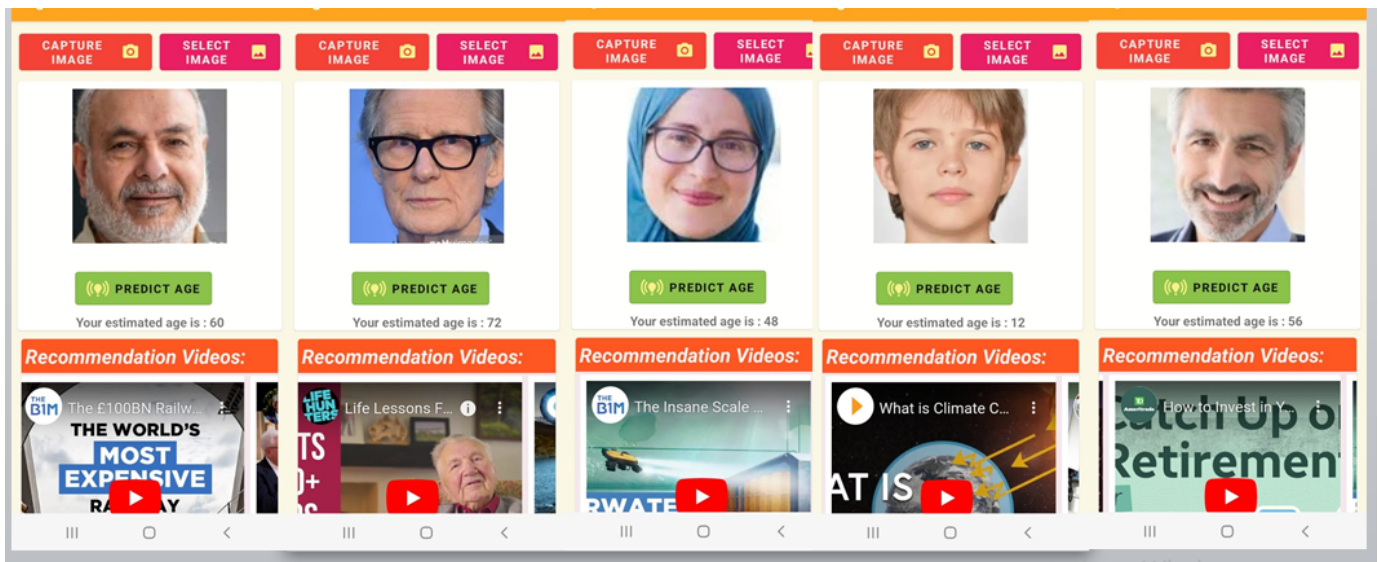


Fig. 9. System implementation of 5 different aged image

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